NINE STRUCTURAL FACTS AROUND THE NORMALIZATION OF A REDUCED NAGATA SCHEME OF DIMENSION 1

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof explicitly reuses the delivered normalization lemma for the stalk statements and the delivered completion lemma for assertions (6)-(9); the short exact sequence and the length identification are carried out in the span.

Allowed standard prerequisites (editorial).

- Normalization of a Noetherian dimension-1 scheme. delivered companion lemma-normalize-noetherian-dim-1. Source of the stalk descriptions and the product-of-Dedekind structure.
- Finiteness of normalization for Nagata schemes. Morphisms, Lemma morphisms-lemma-nagata-normalization. Finiteness of nu.
- The regular locus of a reduced Nagata scheme is dense. More on Algebra, Proposition more-algebra-proposition-ubiquity-J-2. Density of the isomorphism locus of nu.
- Delta invariant vanishes exactly at regular points. delivered companion lemma-delta-invariant-is-zero. Singular-point dichotomy for Q_x.
- Integral closures after henselization and completion. delivered companion lemma-normalization-same-after-completion. Assertions (6)-(9) on A-hat and (A')-hat.

Source reference. Original lemma 0C45: sources/stacks-chow/tags.txt, source lines 15215–15215. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 035S: sources/stacks-chow/tags.txt, source lines 4139–4139. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07PJ: sources/stacks-chow/tags.txt, source lines 9799–9799. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0C3U: sources/stacks-chow/tags.txt, source lines 15204–15204. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0C3V: sources/stacks-chow/tags.txt, source lines 15205–15205. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a reduced Nagata scheme of dimension 1. Let $\nu : X^\nu \to X$ be the normalization. Let $x \in X$ denote a closed point. Then*

(1) *$\nu : X^\nu \to X$ is finite, surjective, and birational,*

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(2) $\mathcal{O}_X \subset \nu_*\mathcal{O}_{X^\nu}$ *and* $\nu_*\mathcal{O}_{X^\nu}/\mathcal{O}_X$ *is a direct sum of skyscraper sheaves with value* $\mathcal{Q}_x$ *at* x *which is nonzero if and only if* x *is a singular point of* X*,*
(3) $A' = (\nu_*\mathcal{O}_{X^\nu})_x$ *is the integral closure of* $A = \mathcal{O}_{X,x}$ *in its total ring of fractions,*
(4) $\mathcal{Q}_x = A'/A$ *has finite length equal to the* δ*-invariant of* X *at* x*,*
(5) A' *is a semi-local ring which is a finite product of Dedekind domains,*
(6) $A^\wedge$ *is a reduced Noetherian complete local ring of dimension* 1*,*
(7) $(A')^\wedge$ *is the integral closure of* $A^\wedge$ *in its total ring of fractions,*
(8) $(A')^\wedge$ *is a finite product of complete discrete valuation rings, and*
(9) $A'/A \cong (A')^\wedge/A^\wedge$.

Proof. We may and will use all the results of Lemma 0C45. Finiteness of ν follows from Morphisms, Lemma 035S. Since X is reduced, Nagata, of dimension 1, we see that the regular locus is a dense open $U \subset X$ by More on Algebra, Proposition 07PJ. Since a regular scheme is normal, this shows that ν is an isomorphism over U. Since $\dim(X) \leq 1$ this implies that ν is not an isomorphism over a discrete set of closed points $x \in X$. In particular we see that we have a short exact sequence

$$0 \to \mathcal{O}_X \to \nu_*\mathcal{O}_{X^\nu} \to \bigoplus\nolimits_{x \in X \setminus U} \mathcal{Q}_x \to 0$$

As we have the description of the stalks of $\nu_*\mathcal{O}_{X^\nu}$ by Lemma 0C45, we conclude that $Q_x = A'/A$ indeed has length equal to the δ-invariant of X at x. Note that $Q_x \neq 0$ exactly when x is a singular point for example by Lemma 0C3U. The description of A' as a product of semi-local Dedekind domains follows from Lemma 0C45 as well. The relationship between A, A', and $(A')^\wedge$ we have see in Lemma 0C3V (and its proof). □